Research Log Book

Aniket Sarrin

Project: Knowledge Tracing for Large Language Models

CMPE 295A, San José State University, Fall 2026 · Advisor: Dr. Mahima Suresh Agumbe

Team: Ronak Patel, Bhagyesh Rathi, Aniket Sarrin · Repository: github.com/bhagyeshrathi07/LLM-Knowledge-Tracing

Period covered: August 31 – October 29, 2026 · Prepared October 29, 2026

Role and track

Data analyses, statistical tests and Phase 2 evaluation (track owner since Sep 29); psychometrics-for-LLMs and checkpoint-release reading; meeting notes.

Summary of contribution to date

- Wrote the one-paragraph problem statement that seeded the abstract.
- Read and summarised the psychometrics-for-LLMs and checkpoint-release literature (Fluid Benchmarking, MathCAMPS, Pythia, OLMo); surveyed which checkpoint releases publish item-level results and read the Fluid Benchmarking data section, which led to the dataset the project uses.
- Drafted the repository structure and the canonical data schema (model, checkpoint, item, skill, correct); created the meetings folder and template; note-taker for the Sep 8 and Sep 15 team meetings.
- Built the forgetting analysis: the exact McNemar peak-to-final test with Holm and selection adjustment (src/llm_kt/forgetting.py, scripts/run_forgetting.py), and the chance-adjusted check that recovers answer-letter preference from gold letters; showed that none of the forgetting candidates survives and that four of the five surviving pairs are letter artefacts.
- Built the static IRT persistence baseline (src/llm_kt/static_irt.py: Fluid 2PL parameter loader, per-skill ability fit, carry-forward predictor).
- Filled the tokens-seen clock for Pythia, Amber and K2 (src/llm_kt/time_metadata.py) and wrote the Phase 2 compute plan with GPU options and a first cost estimate (docs/phase2-gpu-plan.md); owns the Phase 2 pilot (Nov 25), the go/no-go memo (Dec 12) and the advisor's held-out transfer test (Nov 2).

Weekly log

Each entry lists the meetings attended, the work done that week by this member (as opposed to the team), and the artifacts that evidence it (repository paths, commits, documents).

Week 1: Aug 31 – Sep 6

Meetings	Aug 31 advisor meeting (Google Meet).
Work done	• Took part in choosing the project from the advisor's proposed ideas and refining the framing. • Sep 1: wrote the one-paragraph problem statement that seeded the abstract (action item, due Sep 1). • Took the action items to collect and read the papers the advisor shares and to survey the Pythia, OLMo and LLM360 releases for item-level evaluation logs (both due Sep 14).
Artifacts	meetings/2026-08-31-advisor.md; problem statement (abstract draft).

Week 2: Sep 7 – Sep 13

Meetings	Sep 8 team meeting (Discord; note-taker).
Work done	• Wrote the Sep 8 meeting note. • Took the psychometrics-for-LLMs and checkpoint-release reading: Fluid Benchmarking (Hofmann et al. 2025), MathCAMPS (Mishra, Poesia & Goodman 2025), Pythia (Biderman et al. 2023), OLMo (Groeneveld et al. 2024); wrote one-paragraph summaries. • Surveyed which checkpoint releases publish item-level results (Pythia, OLMo, LLM360).
Artifacts	meetings/2026-09-08-team.md; paper summaries.

Week 3: Sep 14 – Sep 20

Meetings	Sep 15 team meeting (Discord; note-taker).
Work done	• Wrote the Sep 15 meeting note. • Presented the summaries: Fluid Benchmarking fits static 2PL IRT per checkpoint and releases item parameters; MathCAMPS gives 44 Common Core skills but reports only accuracy curves. Identified AI2's Fluid Benchmarking release (six open models, hundreds of checkpoints, six benchmarks) as the candidate Phase 1 data source. • Took the action item to read the Fluid Benchmarking paper's data section and README and note what the release contains (due Sep 22).
Artifacts	meetings/2026-09-15-team.md.

Week 4: Sep 21 – Sep 27

Meetings	Sep 23 team meeting (Discord).
Work done	• Reported on the Fluid Benchmarking README: it lists IRT models, accuracy scores and ability estimates but does not say explicitly whether per-item correctness per checkpoint is included; the team agreed to verify by downloading the files. • Took the action items to draft the repository structure and canonical data schema, and to create the meetings folder and template (both due Sep 29).
Artifacts	meetings/2026-09-23-team.md.

Week 5: Sep 28 – Oct 4

Meetings	Sep 29 team meeting (Discord). Workbook 1 submitted Oct 1.
Work done	• Delivered the draft repository structure and the canonical data schema (model, checkpoint, item, skill, correct), which the ingest pipeline implements as the long-format Parquet table. • Created the meetings folder layout and the meeting-note template (meetings/TEMPLATE.md). • Took part in adopting Decisions 1 to 5 and the three-track split (Aniket: data analyses, statistical tests, Phase 2 evaluation); wrote the individual-contribution paragraph for Workbook 1. • Took the Sprint 3 action items: flip-rate and McNemar paired test with selection control on the seven candidate subjects (due Oct 7); fill tokens_seen for Pythia and Amber (due Oct 7); identify GPU options and rough cost for Phase 2 (due Oct 6).
Artifacts	meetings/TEMPLATE.md, meetings/README.md; docs/workbook/Workbook1.pdf §9.1.

Week 6: Oct 5 – Oct 11

Meetings	None recorded.
Work done	• Wrote src/llm_kt/forgetting.py: the exact one-sided McNemar test on items that flip right-to-wrong versus wrong-to-right between a subject's peak checkpoint and its last, run for every (model, subject) pair, with Holm adjustment across the 57 subjects, an additional adjustment for choosing the peak after looking, and a minimum-peak-index guard; scripts/run_forgetting.py with --min-peak-idx; tests in tests/test_forgetting.py. Result: the seven OLMo-2 candidates have raw p-values 0.002 to 0.049 and adjusted p-values of 1.0; five (model, subject) pairs across all models survive, led by OLMo-1 moral_scenarios (0.37 to 0.24, n = 895). • Wrote src/llm_kt/static_irt.py: loader for Fluid's 2PL item parameters, logistic probability, per-skill maximum-a-posteriori ability fitted on a window of history, and the persistence predictor that carries the ability forward; a diagnose() helper for held-out-item checks; tests in tests/test_static_irt.py. Scored on OLMo-2 MMLU as the psychometric baseline (test log-loss 0.393 before calibration).
Artifacts	src/llm_kt/forgetting.py, static_irt.py; tests/test_forgetting.py, test_static_irt.py; scripts/run_forgetting.py; data/processed/forgetting_peak_to_final.parquet.

Week 7: Oct 12 – Oct 18

Meetings	None recorded.
Work done	• Extended the forgetting run and the static IRT baseline for the horizon study: results at h = 5, 10 and 20 feed Chapter 6 of Workbook 2; the recalibrated static IRT (test log-loss 0.393) became the input to the calibration-layer comparison (0.382 after calibration). • Static IRT diagnosis variant (ability fitted on part of a checkpoint's items, scored on the rest) deferred to Sprint 5, where it joins the held-out transfer test.
Artifacts	data/processed/models_mmlu_*.parquet (static_irt rows).

Week 8: Oct 19 – Oct 25

Meetings	Oct 19 advisor meeting (11:20–11:40 PST).
Work done	• Took the action items from the advisor meeting: chance-adjusted forgetting check (answer-letter distribution at peak vs final; due Oct 26) and the design and run of the advisor's held-out test of fitted trajectories (due Nov 2). • Oct 20: built the chance-adjusted forgetting check. Fluid stores only correctness, so the model's answer-letter preference is recovered indirectly: accuracy on items whose gold answer is a given letter equals how often a guessing model picks that letter. Matched gold letters for all 14,042 MMLU items from the public test split (match_mmlu_gold), computed per-letter accuracy at peak and final, a G-test for a change in letter preference, the letter-expected accuracy and the residual drop (letter_check), plus correctness-only nulls: chance proximity with Wilson intervals, the drop's percentile among the other 56 subjects, rest-of-MMLU flip rates (chance_check). Added --chance-check to scripts/run_forgetting.py; 32 tests in tests/test_forgetting.py. • Verdicts: four of the five surviving pairs (Pythia-2.8B and Pythia-6.9B security_studies, Pythia-6.9B and Amber professional_medicine) answered 'D' to essentially every item at their peak and shifted letter later; the letter accounts for 0.17 to 0.26 of each drop, residual at most 0.015. OLMo-1 moral_scenarios is a one-checkpoint spike over an always-'A' model. No forgetting claim survives on Phase 1 data. Merged with PR #3 on Oct 27.
Artifacts	src/llm_kt/forgetting.py (chance_check, letter_check, match_mmlu_gold); scripts/run_forgetting.py; tests/test_forgetting.py; data/processed/forgetting_chance_check.parquet; meetings/2026-10-19-advisor.md.

Week 9: Oct 26 – Oct 29

Meetings	No meeting.
Work done	• Oct 28: filled the tokens-seen clock (src/llm_kt/time_metadata.py, 17 tests; --no-fill-tokens opt-out in scripts/ingest_fluid.py): Pythia derived exactly from 2,097,152 tokens per step (143,000 steps = 300B); Amber estimated from 360 equally spaced checkpoints over 1,259B tokens; K2 estimated from 360 Stage-1 checkpoints over 1.3T. Every checkpoint now carries a token count and a tokens_seen_source of published, derived or estimated; no nulls remain in the 12.6-million-row table. • Oct 28: wrote docs/phase2-gpu-plan.md: what Phase 2 must compute (50 to 100 OLMo-2 checkpoints x 9,607 MathCAMPS problems, generation-based), throughput assumptions per GPU class, options with October 28 prices (SJSU CoE HPC by faculty request, Colab Pro/Pro+, Lambda, RunPod, Vast.ai), a cost range of about $30 to $170 for the full run, the pilot protocol for Nov 25 and the go/no-go rule for Dec 12. • Held-out transfer test of fitted trajectories: design agreed for Sprint 5 (fit on half the items per subject and on MMLU only; score the same checkpoints on the held-out items and on ARC and HellaSwag), due Nov 2.
Artifacts	src/llm_kt/time_metadata.py; tests/test_time_metadata.py; scripts/ingest_fluid.py; docs/phase2-gpu-plan.md (pushed on aniket_branch).

Signed: _____ Date: _____